

Electronic Store Inventory Management System

Software Requirements Specification (SRS) Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	System shall allow authorized users to add, update, and delete product records (name, SKU, category, price, quantity, supplier)	High	Product CRUD operations succeed with valid input; duplicate SKU rejected	Accurate product data is the foundation for all inventory operations
FR-002	Functional	System shall automatically update stock quantity on sale, return, or restock	High	Stock count reflects correct value within 2 sec of transaction	Prevents overselling and manual reconciliation errors
FR-003	Functional	System shall generate an alert when stock falls below a predefined reorder threshold	High	Alert triggers immediately when threshold condition met	Enables timely restocking and avoids stockouts
FR-004	Functional	System shall allow staff to create a sale transaction and	High	Invoice generated with itemized list,	Core business function; required for customer

		generate an invoice		total, date, transaction ID	transactions and record-keeping
FR-005	Functional	System shall authenticate users and restrict actions based on role (Admin, Manager, Staff)	High	Unauthorized role access returns "Access Denied"	Protects sensitive data and prevents unauthorized actions
FR-006	Functional	System shall allow admins to add/edit supplier details and link products to suppliers	Medium	Supplier record saved; linked products display correct supplier	Enables efficient reordering and supply chain tracking
FR-007	Functional	System shall allow managers to create purchase orders and mark them as received	High	PO created with "Pending" status; stock updates on "Received"	Supports systematic restocking workflow
FR-008	Functional	System shall allow users to search and filter products by name, SKU, category, or stock status	Medium	Search returns correct matching results within 1 sec	Improves usability during high-volume checkout or lookup
FR-009	Functional	System shall generate sales, stock, and revenue reports over a selectable date range	Medium	Report totals match underlying transaction records for selected range	Supports business decision-making and performance tracking

FR-010	Functional	System shall log key actions (stock edits, deletions, price changes) with timestamp and user ID	High	Every CRUD action creates a corresponding audit log entry	Ensures accountability and supports dispute resolution
FR-011	Functional	System shall allow staff to process returns/refunds, updating stock and transaction records	Medium	Returned item quantity added back to stock; refund recorded	Supports accurate inventory and financial reconciliation
FR-012	Functional	System shall allow password reset for users who are locked out of their accounts	Low	User receives reset link/OTP and can set a new password successfully	Reduces support overhead and downtime due to lost credentials
NFR-001	Non-Functional	System shall process a sale transaction and update inventory within 2 seconds	High	Transaction completion time ≤ 2 sec under normal load	Ensures smooth checkout experience, avoids customer wait times
NFR-002	Non-Functional	System shall maintain 99% uptime during store operating hours	High	Uptime monitoring logs show $\geq 99\%$ availability	Store operations depend on continuous system access

NFR-003	Non-Functional	Passwords shall be stored using secure hashing (e.g., bcrypt); RBAC enforced on all endpoints	High	No plaintext passwords in DB; restricted endpoints return 401/403 for unauthorized calls	Protects against data breaches and unauthorized access
NFR-004	Non-Functional	UI shall allow a new staff member to complete a basic sale within 15 minutes of training	Medium	New user completes sale transaction unaided within 15 min	Reduces onboarding cost and training overhead
NFR-005	Non-Functional	System shall ensure atomic transactions so stock and billing records never desynchronize	High	No partial updates occur even if a transaction fails mid-way	Maintains data consistency and financial accuracy
NFR-006	Non-Functional	System shall support up to 10,000 product records and 50 concurrent users without performance degradation	Medium	Load test with 50 concurrent users shows no crash or major slowdown	Ensures system scales with business growth

NFR-007	Non-Functional	System shall be accessible via modern web browsers (Chrome, Firefox, Edge) without platform-specific installation	Medium	UI renders and functions correctly across all three browsers	Ensures accessibility across different store devices
NFR-008	Non-Functional	Codebase shall follow modular architecture (MVC) to allow independent module updates	Medium	Inventory, billing, and reporting modules can be updated independently without breaking others	Improves long-term maintainability and reduces regression risk
NFR-009	Non-Functional	System shall perform automated daily database backups with a 24-hour recovery point objective	High	Backup file generated daily and successfully restorable within 24 hrs	Protects against data loss and supports business continuity
NFR-010	Non-Functional	System shall retain audit logs for a minimum of 6 months	Medium	Logs older than current date but within 6 months remain queryable	Supports compliance and long-term traceability
NFR-011	Non-Functional	System shall encrypt sensitive data (e.g., user credentials) in transit using HTTPS/TLS	High	All API/browser communication occurs over HTTPS; no plaintext	Prevents interception of sensitive data over the network

				transmission detected	
NFR-012	Non-Functional	System shall provide clear, consistent error messages for all failed operations	Low	Every failure case (invalid input, network error, etc.) shows a specific, user-understandable message	Improves usability and reduces user confusion/support tickets